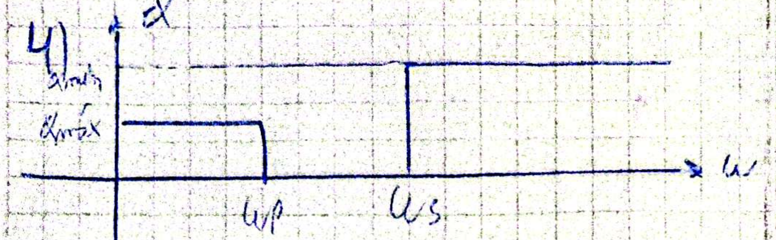
4)



- Pass Band $0 \leq w \leq w_P$
- Transition Band $w_P \leq w \leq w_S$
- Stop Band $w_S \leq w$

- $\alpha_{máx} = 0{,}4\,dB$; $w_P = 2\pi \cdot 3{,}2KHz$; $f_P = f_c = 3{,}2\,KHz$
- $\alpha_{mín} = 48\,dB$, $w_S = 2\pi \cdot 9{,}6KHz$; $f_S = 9{,}6KHz$
- $N_\Omega = W_P[\frac{\sin}{(rad/s)}] = 3.200$

$\Omega_N = 3200 \cdot 2\pi$

$\Rightarrow \Omega_{cN} = \frac{w_P = 2\pi\, 3.200}{N_\Omega\ 2\pi\, 3.200} = 1\ \Omega_{rad/s}$

- $\Rightarrow \Omega_{sN} = \frac{w_S}{N_\Omega} = 3\ \Omega_{rad/s}$

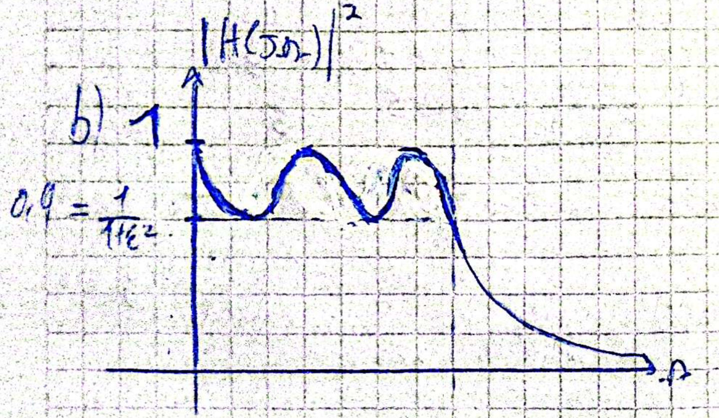
a)

$$m \simeq \frac{\ln\sqrt{4\cdot(10^{0{,}1\cdot\alpha_{mín}}-1)/(10^{0{,}1\cdot\alpha_{máx}}-1)}}{\ln\left(\Omega_{sN}+\sqrt{\Omega_{sN}^2-1}\right)} \approx 4{,}2 \simeq 5 \rightarrow \text{cant de ceros de } |H(j\Omega)|^2$$

$\rightarrow$ Normalizado

$$\varepsilon^2 = 10^{\alpha_{máx}\cdot 0{,}1} - 1 \Rightarrow \varepsilon = 0{,}31$$

b)



- $m = 5 \rightarrow$ Impar $\rightarrow |H(0)| = 1$
- $(m-1) \rightarrow$ picos

c) $$a = \frac{1}{m}\ln\left(\frac{1}{\varepsilon}+\sqrt{\frac{1}{\varepsilon^2}+1}\right) = 0{,}378$$

$$\sigma_K = -\operatorname{sen}\left(\left(\frac{2K-1}{2m}\right)\cdot\pi\right)\cdot \operatorname{senh}(a)$$
$\rightarrow K=1 \Rightarrow \sigma_{1,2} = -0{,}12$
$\rightarrow K=2 \Rightarrow \sigma_{3,4} = -0{,}3131$
$\rightarrow K=3 \Rightarrow \sigma_5 = -0{,}387$

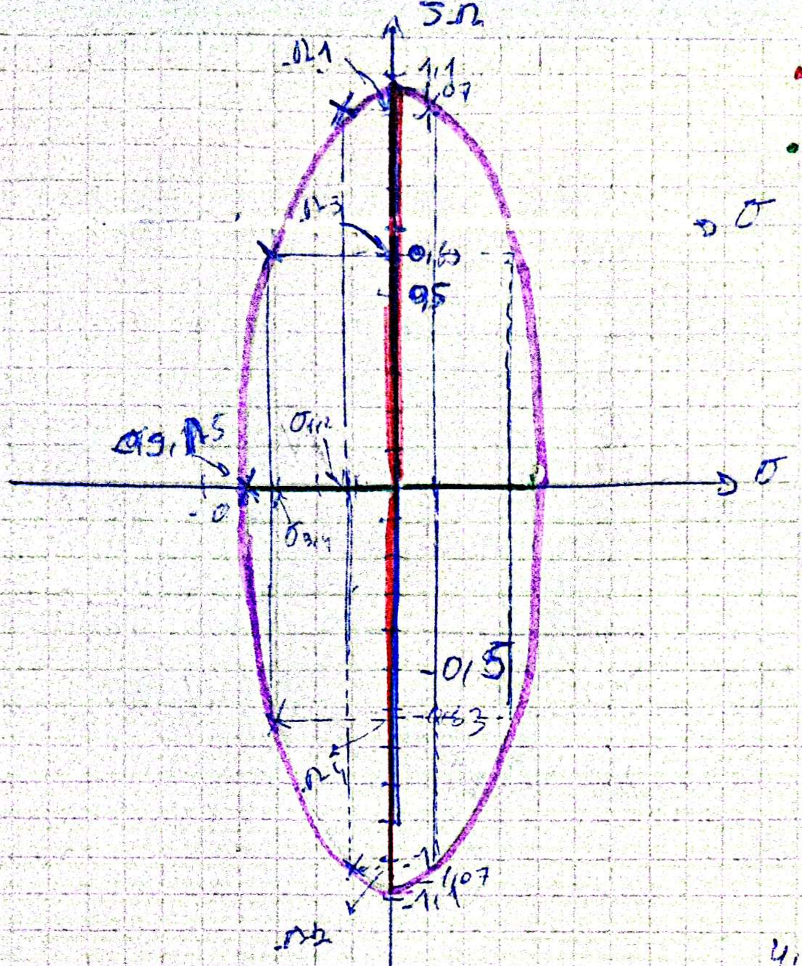
$$\Omega_K = \cos\left(\left(\frac{2K-1}{2m}\right)\pi\right)\cosh(a)$$
$\rightarrow K=1 \Rightarrow \Omega_{1,2} = \pm 1{,}02$
$\rightarrow K=2 \Rightarrow \Omega_{3,4} = \pm 0{,}63$
$\rightarrow K=3 \Rightarrow \Omega_5 = 0$

$\operatorname{senh}(a) = \frac{e^a - e^{-a}}{2}$ $\quad \operatorname{senh}(a)^{-1} = \ln\left(a+\sqrt{a^2+1}\right)$

$\cosh(a) = \frac{e^a + e^{-a}}{2}$ $\quad \cosh(a)^{-1} = \ln\left(a+\sqrt{a^2-1}\right),\ a \geq 1$

$\cosh(\varepsilon) = 1{,}07$

$\operatorname{senh}(\varepsilon) = 0{,}387$



- Factor Q de cada polo $Q_{0K} = \frac{\Omega_{0K}}{\sigma_K \cdot 2}$ → $4{,}28 = Q_{0_{1,2}}$; $1{,}13 = Q_{0_{3,4}}$; $0{,}5 = Q_5$

- Distancia de cada polo al origen ⟹ $\Omega_{0K} = \sqrt{\sigma_K^2 + \Omega_K^2}$ → $\Omega_{0_{1,2}} = 1{,}027$; $\Omega_{0_{3,4}} = 0{,}702$; $\Omega_{0_5} = 0{,}38$

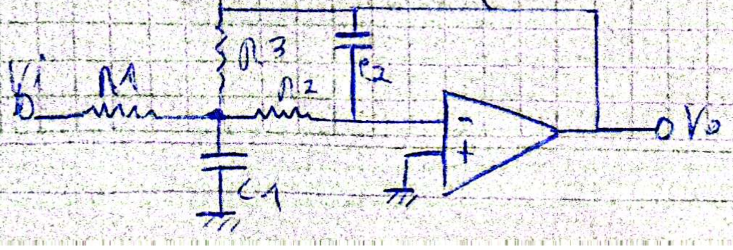
- Fase de cada par de polos $\theta_K = \operatorname{arctg}\left(\frac{\Omega_K}{\sigma_K}\right) =$ $\theta_{1,2} = \pm 83{,}3^\circ$; $\theta_{3,4} = \pm 63{,}6^\circ$; $\theta_5 = 0^\circ$

d) Síntesis por Multiple Feedback:

- Planteo transferencia de FPB de orden 2: $H(s) = \frac{W_0^2}{s^2 + \frac{W_0}{Q}s + W_0^2}$ con $W_0 = \Omega_{0K}$

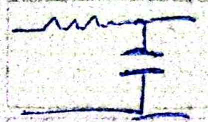
- Planteo transferencia de Filtro OPAMP MFB orden 2:

$$H(s) = \frac{-1/R_1 R_2 C_1 C_2}{s^2 + s\,\frac{1}{C_1}\left(\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}\right) + \frac{1}{R_2 R_3 C_1 C_2}}$$



NOTA

Planteo transferencia con RC. TPB orden n=1

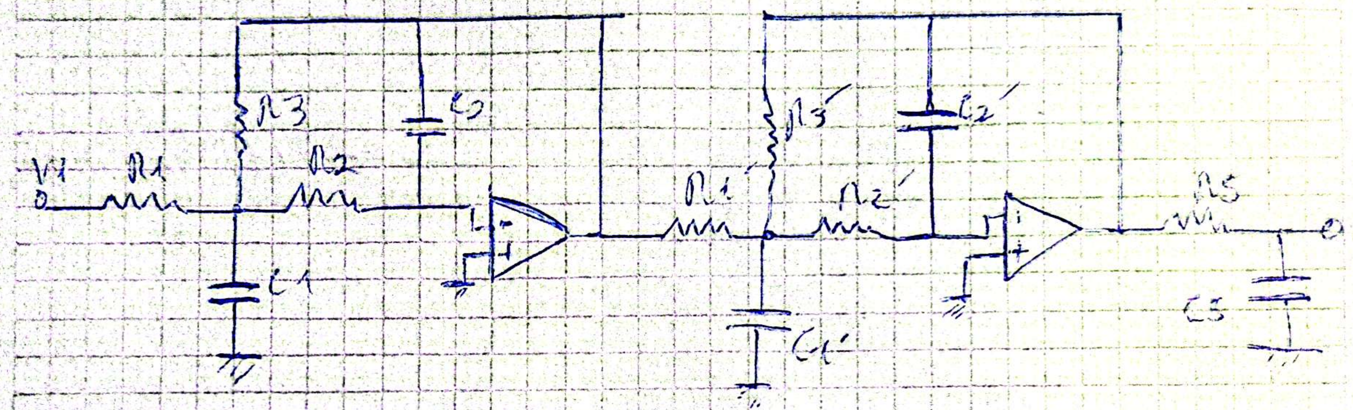


$$H(\$) = \frac{\omega_0}{\$+\omega_0} , \quad \omega_0 = \frac{1}{R \cdot C}$$

Por inspección entre ambas transferencias queda:

$$\Omega_{0k}^2 = \frac{1}{R_1 \cdot R_2 \cdot C_1 \cdot C_2} , \quad \frac{\Omega_{0k}}{Q_{0k}} = \frac{1}{C_1}\left(\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}\right) , \quad \Omega_{0k} = \frac{1}{R \cdot C}$$

Circuito:



• De (1) $\boxed{R_{1,2,3N} = 1\Omega , \ Z_N = 1K}$ por criterio de diseño

$\Rightarrow$ Pongo polos $P_{1/2}(\Omega_{0\,1,2} ; Q_{1,2}) \Rightarrow \frac{\Omega_{0\,1,2}}{Q_{0\,1,2}} = \frac{1}{C_{1N}}\left(\frac{1}{R_{1N}} + \frac{1}{R_{2N}} + \frac{1}{R_{3N}}\right)$

$\Rightarrow \frac{1{,}027}{4{,}123} = \frac{1}{C_{1N}}\left(\frac{1}{1} + \frac{1}{1} + \frac{1}{1}\right) \Rightarrow \boxed{C_{1N} = 12{,}3\,F}$

y aplicando $\Omega_{0\,1,2}^2 = \frac{1}{R_{1N} R_{2N} C_{1N} C_{2N}} \Rightarrow \boxed{C_{2N} = 0{,}0758\,F}$

Desnormalizando: (con $\Omega_N = 3200 \cdot 2\pi$

$C_1 = \frac{C_{1N}}{Z_N \cdot \Omega_N} = \boxed{621{,}7\,mF} \rightarrow C_1$; $C_2 = \frac{C_{2N}}{Z_N \cdot \Omega_N} = \boxed{3{,}77\,mF} \rightarrow C_2$

$R_{1,2,3} = R_{1,2,3N} \cdot Z_N = \boxed{1K\Omega = R_{1,2,3}}$

• De ② $\boxed{R'_{1,2,3N} = 1\Omega \text{ y } Z_N = 1K}$ por criterio de diseño

=> Por ende polos $P_{3,4} = (\sigma_{3,4}; \Omega_{3,4})$ => $\frac{\Omega_{03,4}}{Q_{03,4}} = \frac{1}{C1N'}\left(\frac{1}{R1N'}+\frac{1}{R2N'}+\frac{1}{R3N'}\right)$

=> $\frac{0,702}{1,13} = \frac{1}{C1N'}\cdot\left(\frac{1}{1}+\frac{1}{1}+\frac{1}{1}\right)$ => $\boxed{C1N' = 4,83\,F}$

y aplicando $\Omega_{03,4}^2 = \frac{1}{R1N'\,R2N'\,C1N'\,C2N'}$ => $\boxed{C2N' = 0,42\,F}$

Desnormalizando con $\Omega N = 3200\cdot 2\pi$

$C1' = \frac{C1N'}{Z_N\cdot\Omega_N} = \boxed{240,22\,mF} \to C1'$, $C2' = \frac{C2N'}{Z_N\cdot\Omega_N} = \boxed{20,9\,mF} \to C2'$

$R'_{1,2,3} = R_N'\cdot Z_N = \boxed{1K\Omega = R'_{1,2,3}}$

• De ③ $\boxed{R5N = 1\Omega \text{ y } Z_N = 1K}$ por criterio de diseño

=> Por ende 1 Polo $P_5 = (\sigma_5, \Omega_5)$ => $\Omega_{05} = \frac{1}{R_{5N}\cdot C_{5N}}$

=> $0,38 = \frac{1}{1\Omega\cdot C5N}$ => $\boxed{C5N = 2,631\,F}$

Desnormalizando con $\Omega N = 3200\cdot 2\pi$

$C5 = \frac{C5N}{Z_N\,\Omega_N} = \boxed{130,8\,mF} \to C5$, $R5 = R5N\cdot Z_N = \boxed{1K = R5}$

Simulando vemos que:

$\alpha_{min} > 48\,dB$ con $\alpha_{min} = 60\,dB$ y además más aún de los pedidos en la banda de stop, lo cual es bueno.

$\alpha_{max} > 0,40\,dB$ con $\alpha_{max} = 0,60\,dB$ y además ~~los~~ apenas es más aún en la banda de paso, lo cual es malo pero al ser unos 0,15 dB de diferencia no es tan grave.

NOTA